

MEMMINGEN, Germany

WMO: 109470

Lat: 47.989N

Lon: 10.240E

Elev: 2064

StdP: 13.63

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	13.9	8.5	9.2	10.3	12.1	11.0	14.6	30.5	46.0	27.4	41.6	3.5	60	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.3	84.2	65.7	80.7	64.3	77.5	62.7	66.8	81.0	65.3	78.7	63.8	75.7	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	87.2	72.6	60.0	83.4	71.1	58.7	79.5	69.6	32.5	81.3	31.2	78.8	30.1	76.0	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.3	20.4	17.2	2.0	89.3	3.9	2.7	-0.8	91.3	-3.1	92.9	-5.3	94.4	-8.2	96.4		
(5)			1.9	69.4	3.9	1.5	-0.9	70.5	-3.1	71.4	-5.3	72.2	-8.1	73.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	31.0	34.0	40.3	44.9	54.7	59.3	63.3	63.9	55.6	47.9	37.7	32.2
	DBStd	13.59	8.25	8.77	7.14	7.08	6.57	6.53	5.90	5.89	5.53	7.21	7.17	8.04
	HDD50	2627	588	449	309	185	30	4	0	1	12	124	371	554
	HDD65	6655	1053	868	767	603	325	195	103	91	283	530	819	1018
	CDD50	1585	0	1	7	33	174	282	413	431	182	60	2	0
	CDD65	136	0	0	0	0	4	23	50	57	2	0	0	0
	CDH74	1589	0	0	0	3	97	278	540	627	43	1	0	0
	CDH80	355	0	0	0	0	8	50	140	155	2	0	0	0
	WSAvg	6.6	7.3	7.6	8.1	6.9	6.0	6.1	6.0	5.2	5.7	6.5	6.6	7.7
	PrecAvg	48.7	3.0	2.9	3.3	3.3	4.9	5.7	5.6	5.2	3.9	3.5	3.7	3.7
(15) Precipitation	PrecMax	63.9	6.8	6.7	8.0	7.2	9.8	9.3	11.4	10.2	7.1	7.0	9.1	7.6
	PrecMin	39.1	0.2	1.1	1.1	0.6	0.6	1.7	1.0	2.1	1.8	0.8	0.3	0.6
	PrecStd	6.2	1.6	1.5	1.6	1.7	2.2	1.9	2.4	2.2	1.6	1.6	1.9	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	60.6	64.6	73.5	80.9	85.5	88.1	88.7	78.7	71.7	58.9	53.2
		MCWB	44.6	50.0	51.5	54.8	60.5	64.9	67.9	66.3	63.2	59.4	51.4	46.5
	2%	DB	48.5	54.6	59.9	68.3	76.7	80.6	84.1	84.3	74.6	66.5	53.4	48.5
		MCWB	43.0	45.6	49.3	54.0	59.8	64.0	65.9	65.7	61.4	57.2	47.6	43.9
	5%	DB	45.0	49.6	55.5	62.9	72.8	77.1	80.1	80.7	71.1	62.3	49.9	46.1
		MCWB	40.7	42.6	47.2	51.4	58.1	62.1	64.9	64.4	60.1	55.1	45.7	42.5
	10%	DB	42.4	45.6	51.6	58.5	68.8	73.1	76.2	77.3	66.5	58.3	47.8	43.3
		MCWB	38.9	40.6	45.0	48.7	56.2	60.4	62.9	63.0	57.7	53.1	44.5	40.5
	0.4%	WB	45.3	49.3	53.0	56.7	62.8	67.5	68.8	68.1	64.7	61.4	51.9	47.6
		MCDB	51.1	58.7	63.3	69.1	76.3	79.9	84.8	82.8	76.6	69.3	57.8	52.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.3	46.0	50.0	54.4	60.7	65.0	66.9	66.6	62.2	58.1	48.4	44.3
		MCDB	47.4	53.2	58.1	66.5	74.7	78.1	81.4	81.2	72.4	64.6	51.7	47.6
	5%	WB	41.2	43.2	47.8	52.1	58.8	63.0	65.4	65.2	60.5	55.6	46.6	42.6
		MCDB	44.8	49.0	54.6	62.8	71.2	75.5	78.6	78.8	69.7	60.9	49.9	45.6
	10%	WB	38.8	40.7	45.5	49.3	56.7	61.0	63.6	63.7	58.4	53.5	44.3	40.6
		MCDB	41.5	44.9	50.9	57.2	67.3	71.9	74.6	75.7	65.3	58.2	47.0	43.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	15.1	15.3	17.3	19.6	18.8	18.9	20.3	17.6	14.7	11.5	10.1
		MCDBR	16.2	22.5	22.8	26.5	27.8	26.4	26.5	27.1	25.5	21.8	17.2	13.6
		MCWBR	11.7	15.0	14.3	15.2	14.2	13.2	12.6	12.6	14.5	13.9	12.1	9.9
	5% WB	MCDBR	15.0	20.1	20.8	24.6	25.8	24.1	23.8	24.5	22.5	20.4	15.0	12.5
		MCWBR	11.2	13.9	13.4	14.5	13.6	12.4	11.7	11.9	13.2	13.3	11.0	9.5
(40) Clear-Sky Solar Irradiance	taub		0.287	0.313	0.348	0.378	0.389	0.395	0.396	0.390	0.365	0.339	0.310	0.284
	taud		2.442	2.372	2.343	2.291	2.300	2.315	2.302	2.337	2.386	2.448	2.478	2.453
	Ebn at Noon		253	268	275	277	277	275	273	270	266	256	241	239
	Edh at Noon		21	27	33	38	40	39	39	36	31	25	20	18
(44) All-Sky Solar Radiation	RadAvg		357	632	989	1359	1564	1763	1704	1498	1098	707	388	290
	RadStd		45	86	122	183	175	161	157	154	129	80	52	41

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (92 neighbors)	N/A	N/A	N/A	N/A	+0.76	+0.72	N/A	N/A	+99	+27

Nomenclature: See separate page